

EB-JEPA — TUEV Reference Table (V1)

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Our EB-JEPA runs on **TUEV** (6-class event detection, frozen logistic probe). Companion to `reference_table_JEPA_V1.pdf` (TUAB) and `examples/eeg/PROVENANCE_TABLE_TUEV.md` (literature baselines).

#	Encoder	Params	Energy (SSL objective)	Eval	TUEV BAcc
baseline	Conv1D	0.4M	VICReg — <i>base aug, no corruption</i>	frozen	0.381
1	Conv1D	0.4M	VICReg + spectral 0.1 + corrupt	frozen	0.425 ★ best
2	Conv1D	0.4M	VICReg + corrupt (s0)	frozen	0.382
3	Conv1D	0.4M	SIGReg + corrupt	frozen	0.363

TUEV BAcc — 6-class macro balanced accuracy on TUEV event detection. **Chance** = 0.167; **random-encoder floor** = 0.337. All rows use a **TUAB-pretrained encoder, frozen, never trained on TUEV** — pure representation-transfer probe.

Probe protocol — class-balanced logistic regression on event-centered 5s windows (200 Hz, 19 channels, per-channel z-scored), background class subsampled so the 6 classes (SPSW / GPED / PLED / EYEM / ARTF / BCKG) are balanced. Subject-disjoint train/eval.

Fixed for every corruption run (never swept): views v_1, v_2 made with Gaussian noise $\sigma = 0.1$, scale-jitter $\pm 20\%$, channel-drop $p = 0.2$, plus exact EB-corruption — 20% timepoints masked + 20% $\pm 6\sigma$ outliers ($\approx 40\%$ /view). The **baseline** is the no-corruption reference (legacy contiguous time-mask).

VICReg* note — these runs use the original `inv_coeff = 1` (the 3-seed ablation showed `inv=1` actually beats `inv=25` by ~ 3 pp / $\sim 6\sigma$ on TUAB; reframed as a finding rather than a bug in commit `5a13ebd`).

Reading the table — best frozen-transfer is **row 1 (VICReg + spectral 0.1 + corrupt $\rightarrow$ 0.425 BAcc)**, which beats our supervised CNN-Transformer (0.410) and ST-Transformer (0.339) **without ever seeing a TUEV label**. Every config lands well above the random-encoder floor (0.337) and the 6-class chance line (0.167), confirming the TUAB-pretrained representation transfers to TUEV.